# BCB744 Biostatistics: practical assessment (Version 2)

## Chapters 1–14 | Total: 170 marks

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! Instructions

- Answer **all eight questions**. The paper totals **170 marks** and you have **30 hr** to complete it.
- Submit a thoroughly annotated HTML report rendered from Quarto. Show the R code used and interpret the evidence in complete sentences.
- This is a practical assessment. Marks are awarded for exploratory data analysis, statistical reasoning, correct analysis, diagnostics, interpretation, and communication. No question asks for a stand-alone theory essay.
- Before each inferential analysis, state the biological question and the null and alternative hypotheses. Identify the observational or experimental unit.
- Use plots and numerical summaries to examine the data before choosing or fitting a test. Check assumptions at the level appropriate to the analysis.
- Report estimates or effect sizes, uncertainty where available, test statistics, degrees of freedom, and **exact *p*-values**.
- Unless a question explicitly requests otherwise, use a two-sided significance level of $\alpha = 0.05$.
- Long, unedited console dumps are not acceptable and will be penalised. Present only output that contributes to the analysis.
- **Required YAML Structure:** Your Quarto document must begin with the following YAML header (fill in your own details where indicated):

```yaml
---
title: "BCB744 Biostatistics Assessment"
author: "Your Name"
date: "2026"
format:
  html:
    embed-resources: true
    toc: false
    number-sections: false
---
```

Your document should follow this hierarchical structure:

```
# Question 1

## Preamble
## Introduction
## Methods
## Results
## Discussion

# Question 2

## Preamble
```

## Question 1: Mammalian body and brain mass (/20)

The `mammals` dataset in **MASS** contains body mass (kg) and brain mass (g) for 62 terrestrial mammal species. Each row represents one species.

a. Inspect the structure, missingness, and numerical range of both variables. Produce an informative two-panel EDA figure showing the distribution of raw body mass and raw brain mass. Summarise what the figure and appropriate measures of centre and spread reveal. **(/5)**

b. Apply a base-10 logarithm to both variables. Reproduce the two distribution plots on the transformed scale and explain, using evidence from your plots and summaries, whether the transformation improves their suitability for model-based analysis. **(/5)**

c. Treat the 62 observed species as a finite population. Set the seed to `744`, then draw 2,000 random samples **without replacement** at each of two sample sizes, $n = 10$ and $n = 40$. For every sample, calculate the mean $\log_{10}$ brain mass. Plot the two sampling distributions and report their empirical means, standard errors, and central 95% ranges. Explain what the comparison demonstrates about sample size and sampling uncertainty. **(/6)**

d. Make a scatter plot of $\log_{10}$ brain mass against $\log_{10}$ body mass. Describe the form, direction, strength, and any unusual observations. State one limitation of treating the 62 species as independent replicates. Do not yet fit a regression model. **(/4)**

Worked answer: Question 1

```
mammals_df <- MASS::mammals |>
  tibble::rownames_to_column("species")

str(mammals_df)
```

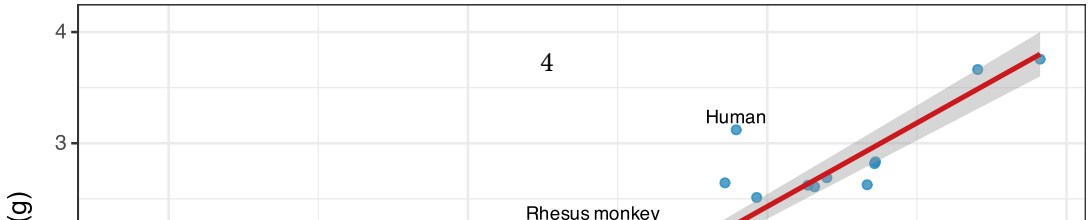
```
'data.frame':   62 obs. of  3 variables:
 $ species: chr  "Arctic fox" "Owl monkey" "Mountain beaver" "Cow" ...
 $ body   : num  3.38 0.48 1.35 465 36.33 ...
 $ brain  : num  44.5 15.5 8.1 423 119.5 ...
```

```
colSums(is.na(mammals_df))
```

```
species    body   brain
      0       0       0
```

```
mammals_df |>
  summarise(
    across(
      c(body, brain),
      list(
        min = min,
        q1 = ~ quantile(.x, 0.25),
        median = median,
        mean = mean,
        q3 = ~ quantile(.x, 0.75),
        max = max,
        sd = sd,
        iqr = IQR
      )
    )
  )
```

```
  body_min body_q1 body_median body_mean body_q3 body_max body_sd body_iqr
1    0.005     0.6      3.3425    198.79 48.2025     6654 899.158  47.6025
  brain_min brain_q1 brain_median brain_mean brain_q3 brain_max brain_sd
1      0.14     4.25        17.25   283.1342      166      5712 930.2789
  brain_iqr
1    161.75
```



## Question 2: A thermal-acclimation experiment (/16)

The following code creates a new dataset from an experiment on 24 individually tagged intertidal snails. Oxygen consumption was measured for each snail before and after a 14-day warm-acclimation treatment. Rates are in µmol $O_2$ $h^{-1}$.

```
snails <- tibble(
  snail_id = sprintf("S%02d", 1:24),
  before = c(
    5.8,
    6.1,
    4.9,
    5.4,
    6.7,
    5.2,
    5.9,
    6.4,
    5.1,
    5.6,
    6.3,
    5.0,
    5.7,
    6.2,
    4.8,
    5.5,
    6.0,
    5.3,
    6.5,
    5.4,
    5.9,
    6.1,
    5.2,
    5.8
  ),
  after = c(
    6.4,
    6.8,
    5.1,
    6.0,
    7.5,
    5.7,
    6.6,
    7.0,
    5.6,
```

```
    6.1,
    7.1,
    5.4,
    6.3,
    6.9,
    5.2,
    6.2,
    6.7,
    5.9,
    7.2,
    5.8,
    6.5,
    6.8,
    5.8,
    6.4
  )
)
```

a. Reshape the data into a tidy long form and produce a graphic that preserves the pairing between measurements. Add a concise numerical summary of the before, after, and within-snail change values. **(/5)**

b. State the experimental unit and the hypotheses. Check the distribution of the **within-snail differences**, explain why this is the relevant diagnostic, and conduct an appropriate inferential test. **(/6)**

c. Report the estimated mean change, its 95% confidence interval, the test statistic, degrees of freedom, and $p$-value. Interpret both the statistical and biological result. **(/5)**

Worked answer: Question 2

```
snails_long <- snails |>
  pivot_longer(c(before, after), names_to = "occasion", values_to = "oxygen")
|>
  mutate(occasion = factor(occasion, levels = c("before", "after")))

snails_change <- snails |>
  mutate(change = after - before)

snails_change |>
  summarise(
    before_mean = mean(before),
    before_sd = sd(before),
    after_mean = mean(after),
    after_sd = sd(after),
    mean_change = mean(change),
    sd_change = sd(change),
    median_change = median(change),
    iqr_change = IQR(change)
  )
```

```
# A tibble: 1 × 8
     before_mean   before_sd   after_mean   after_sd   mean_change   sd_change
median_change
       <dbl>     <dbl>      <dbl>    <dbl>       <dbl>     <dbl>         <dbl>
1        5.7     0.529       6.29    0.648       0.592     0.141           0.6
# ℹ 1 more variable: iqr_change <dbl>
```

```
ggplot(snails_long, aes(occasion, oxygen, group = snail_id)) +
  geom_line(colour = "grey65", alpha = 0.8) +
  geom_point(aes(colour = occasion), size = 2.2, show.legend = FALSE) +
  stat_summary(
    aes(group = 1),
    fun = mean,
    geom = "line",
    colour = "black",
    linewidth = 1.1
  ) +
  stat_summary(
    aes(group = 1),
    fun = mean
```

```
 [1] 0.5920609 0.6512754
attr,"conf.level")
[1] 0.95
data:  snails$after and snails$before
t = 20.951, df = 26, p-value = 2.723e-16
alternative hypothesis: true mean difference is not equal to 0
```

The fitted estimate is the mean **after minus before** change. Its confidence interval excludes zero, and the paired test provides strong evidence against $H_0$. Warm acclimation increased oxygen consumption in this sample by about the reported mean change in μmol $O_2$ $h^{-1}$. The result estimates an average response over 14 days; it does not establish whether the response persists longer or generalises beyond the sampled population.

## Question 3: Bee repellency of orchard sprays (/20)

The built-in `OrchardSprays` dataset records the decrease in the volume of a sugar solution after honeybees were exposed to eight orchard-spray treatments. The decrease is used as an index of bee repellency. There are eight observations per treatment. For this assessment, treat each orchard position as an independent experimental unit and treatment as the sole explanatory factor; do not model `rowpos` or `colpos`.

a. Inspect the data and produce a figure that displays all observations as well as a suitable distributional summary for each treatment. Calculate a suite of suitable summary statistics and report them neatly in a table using the **gt** package. **(/6)**

b. State the experimental unit and hypotheses for an overall treatment comparison. Select an appropriate inferential approach and justify it using the study design, the distribution of model errors, the variability among treatments, and the influence of unusual observations. Show the diagnostic evidence supporting your decision. **(/7)**

c. Conduct the analysis selected in (b). If the overall result justifies further investigation, compare treatment pairs using a procedure that controls the family-wise error rate. Report the overall result and the adjusted pairwise evidence, then explain which treatments differ and in which direction without claiming more than the analysis supports. **(/7)**

Worked answer: Question 3

```
sprays <- as_tibble(OrchardSprays) |>
  mutate(treatment = factor(treatment))

spray_summary <- sprays |>
  summarise(
    n = n(),
    mean = mean(decrease),
    median = median(decrease),
    sd = sd(decrease),
    iqr = IQR(decrease),
    .by = treatment
  ) |>
  arrange(treatment)

spray_summary |>
  gt::gt(rowname_col = "treatment") |>
  gt::tab_header(
    title = "Bee repellency by orchard-spray treatment",
    subtitle = "Decrease in sugar-solution volume"
  ) |>
  gt::cols_label(
    n = "n",
    mean = "Mean",
    median = "Median",
    sd = "SD",
    iqr = "IQR"
  ) |>
  gt::fmt_number(
    columns = c(mean, median, sd, iqr),
    decimals = 2
  )
```

## Bee repellency by orchard-spray treatment

Decrease in sugar-solution volume

| | n | Mean | Median | SD | IQR |
|---|---|---|---|---|---|
| A | 8 | 4.62 | 4.00 | 3.20 | 2.25 |
| B | 8 | 7.62 | 7.50 | 3.29 | 3.00 |
| C | 8 | 25.25 | 16.50 | 24.43 | 7.00 |
| D | 8 | 22.75 | 22.75 | 5.92 | 2.75 |
| E | 8 | 22.75 | 21.50 | 5.92 | 2.75 |
| F | 8 | 19.62 | 19.50 | 3.66 | 3.00 |
| G | 8 | 24.50 | 24.00 | 3.74 | 4.25 |
| H | 8 | 3.50 | 3.50 | 1.51 | 2.25 |

Only pairs with Holm-adjusted $p < 0.05$ should be described as supported differences. Treatments D, E, F, G, and H each had a larger mean decrease than A and B. Treatment G also exceeded C and D, while H exceeded C and D. No other pair had an adjusted $p$-value below 0.05. The omnibus result does not mean every treatment differs from every other treatment, and non-significant pairs should not be declared equivalent.

```
	Pairwise comparisons using t tests with non-pooled SD

data:  sprays$decrease and sprays$treatment

  A       B       C       D       E       F       G
B 1.00000 -       -       -       -       -       -
C 0.00012 0.00012 -       -       -       -       -
D 0.00012 0.00012 1.00000 -       -       -       -
E 0.00012 0.00012 1.00000 1.00000 -       -       -
F 0.00012 0.00012 0.00012 0.00012 0.00012 -       -
G 0.00012 0.00012 0.00012 0.00012 0.00012 1.00000 -
H 1.00000 1.00000 0.00012 0.00012 0.00012 0.00012 0.00012

P value adjustment method: holm
```

```
Kruskal-Wallis rank sum test

data: decrease by treatment
Kruskal-Wallis chi-squared = 40.669, df = 7, p-value = 2.085e-07
```

```
One-way analysis of means (not assuming equal variances)

data: decrease and treatment
F = 33.515, num df = 7.000, denom df = 22.843, p-value = 1.476e-10
```

## Question 4: Allometric scaling of mammalian brains (/22)

Return to the log-transformed mammal data from Question 1. Treat this as a model-building question rather than a repeat of the EDA.

a. Quantify the association between $\log_{10}$ body mass and $\log_{10}$ brain mass. Select and justify an appropriate inferential measure from the form and distribution of the data, state the hypotheses, and report its estimate, 95% confidence interval, test statistic, degrees of freedom, and $p$-value. **(/4)**

b. Fit a suitable model to predict $\log_{10}$ brain mass from $\log_{10}$ body mass. Present the fitted equation, interpret both coefficients and the slope in allometric terms, and report the slope estimate with its 95% confidence interval, test statistic, degrees of freedom, $p$-value, and model $R^2$. **(/6)**

c. Assess whether the fitted model satisfies its distributional and structural assumptions. Quantify case influence, identify the most influential species, and explain whether the evidence warrants deleting any observation. **(/4)**

d. Obtain the mean-response 95% confidence interval and individual-species 95% prediction interval for a mammal weighing 10 kg. Interpret the difference between the two intervals on the original brain-mass scale. **(/4)**

e. Write one concise, publication-style results paragraph integrating the principal numerical findings from (a)–(d). Distinguish the fitted mean response from prediction for a new species and avoid causal language. **(/4)**

Worked answer: Question 4

$H_0 : \rho = 0$ and $H_A : \rho \neq 0$, where $\rho$ is the population Pearson correlation between the two transformed measurements.

```
q4_cor <- cor.test(
  mammals_log$log_body,
  mammals_log$log_brain,
  method = "pearson"
)
q4_cor
```

```

    Pearson's product-moment correlation

data:  mammals_log$log_body and mammals_log$log_brain
t = 26.409, df = 60, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9335522 0.9755352
sample estimates:
      cor
0.9595748
```

The scatter plot from Question 1 supports a linear measure of association on the log scales. The estimated correlation is positive and strong, with its confidence interval well above zero.

```
q4_lm <- lm(log_brain ~ log_body, data = mammals_log)
summary(q4_lm)
```

```

Call:
lm(formula = log_brain ~ log_body, data = mammals_log)

Residuals:
     Min       1Q   Median       3Q      Max
-0.74503 -0.21380 -0.02676  0.18934  0.84613

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.92713    0.04171   22.23   <2e-16 ***
log_body     0.75169    0.02846   26.41   <2e-16 ***
```

```
10^q4_pred      lwr      upr
1 47.73236 31.08098 73.31038
```

Exponentiating converts the fitted values and interval limits back to grams. The confidence interval estimates uncertainty in the **mean** brain mass for the modelled population of 10 kg mammals. The prediction interval is wider because it also includes residual among-species variation and is intended for a single new 10 kg species.

A suitable publication-style result is:

## Question 5: Curvature in an enzyme assay (/18)

The built-in `DNase` dataset contains optical-density measurements from an assay of deoxyribonuclease. Use only observations from `Run == 1`. Treat `conc` as the predictor and `density` as the response.

a. Inspect and plot the relationship. Fit two candidate models for the expected response: the first assumes a constant change in optical density per unit increase in concentration; the second adds the square of concentration as an additional predictor, entered in the original concentration units. Overlay both fitted relationships on the observations and write their equations. **(/6)**

b. Formally determine whether the additional term improves fit sufficiently to retain the more complex candidate. State the hypotheses in terms of that added coefficient and report the comparison statistic, degrees of freedom, and $p$-value. **(/5)**

c. Diagnose the selected model's assumptions and case influence. Interpret the fitted relationship only within the observed concentration range, and use fitted values to quantify the expected optical density at the minimum, median, and maximum observed concentrations. Explain what these estimates reveal about how the response changes across the concentration gradient. **(/7)**

Worked answer: Question 5

```
dnase_1 <- as_tibble(DNase) |>
  filter(Run == 1)

str(dnase_1)
```

```
tibble [16 × 3] (S3: tbl_df/tbl/data.frame)
 $ Run    : Ord.factor w/ 11 levels "10"<"11"<"9"<..: 4 4 4 4 4 4 4 4 4 4 ...
 $ conc   : num [1:16] 0.0488 0.0488 0.1953 0.1953 0.3906 ...
 $ density: num [1:16] 0.017 0.018 0.121 0.124 0.206 0.215 0.377 0.374
0.614 0.609 ...
```

```
range(dnase_1$conc)
```

```
[1]  0.04882812 12.50000000
```

```
q5_linear <- lm(density ~ conc, data = dnase_1)
q5_quadratic <- lm(density ~ conc + I(conc^2), data = dnase_1)

q5_grid <- tibble(
  conc = seq(min(dnase_1$conc), max(dnase_1$conc), length.out = 200)
) |>
  mutate(
    linear = predict(q5_linear, newdata = pick(everything())),
    quadratic = predict(q5_quadratic, newdata = pick(everything()))
  ) |>
    pivot_longer(c(linear, quadratic), names_to = "model", values_to =
"density")

ggplot(dnase_1, aes(conc, density)) +
  geom_point(size = 2.5) +
  geom_line(
    data = q5_grid,
    aes(colour = model),
    linewidth = 1
  ) +
  labs(
    x = "DNase concentration",
    y = "Optical density",
```

```
# A tibble: 3 × 3
  position    conc fitted_density
  <chr>      <dbl>          <dbl>
1 Minimum   0.0488          0.109
2 Median    1.17            0.435
3 Maximum  12.5             1.70
```

The selected model describes a curved increase in optical density over the measured concentration range. The fitted values at the minimum, median, and maximum concentrations quantify the response on the scale of the assay. Their successive changes show that optical density increases rapidly over part of the gradient but that the increase is not constant across the full range. These predictions are interpolations supported by the observed concentrations; the fitted equation should not be used to infer behaviour beyond that range. The diagnostic

## Question 6: Sex, body mass, and heart mass in cats (/24)

The `cats` dataset in **MASS** contains sex (`Sex`), body mass (`Bwt`, kg), and heart mass (`Hwt`, g) for 144 adult cats.

a. Inspect the data and use a scatter plot, group-specific fitted lines, and concise summaries to explore how heart mass varies with body mass and sex. **(/5)**

b. Fit a model that estimates the partial association of body mass and sex with heart mass, initially assuming that the body-mass association has the same slope in both sexes. Write the fitted equation, state term-specific hypotheses, identify the reference sex, interpret every coefficient with units, and report coefficient uncertainty and overall model fit. **(/8)**

c. Compare three candidate models: (i) body mass as the sole predictor; (ii) body mass and sex with a common body-mass slope; and (iii) body mass and sex with the body-mass slope allowed to differ between sexes. Use appropriate formal comparisons to determine what additional structure the data support. Explain why insufficient evidence for different slopes is not proof that the population slopes are identical. **(/5)**

d. For the selected model, assess distributional and structural assumptions, quantify whether the predictors contain redundant information, and identify influential observations. Conclude with a concise results paragraph containing the principal estimates and inferential evidence while distinguishing association from causation. **(/6)**

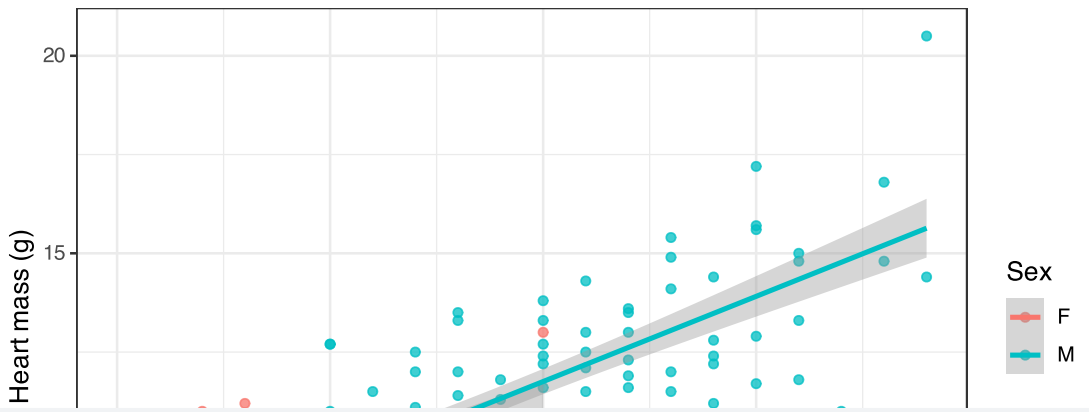
Worked answer: Question 6

```
cats_df <- as_tibble(MASS::cats) |>
  mutate(Sex = relevel(factor(Sex), ref = "F"))

cats_df |>
  summarise(
    n = n(),
    mean_bwt = mean(Bwt),
    sd_bwt = sd(Bwt),
    mean_hwt = mean(Hwt),
    sd_hwt = sd(Hwt),
    .by = Sex
  )
```

```
# A tibble: 2 × 6
  Sex       n mean_bwt sd_bwt mean_hwt sd_hwt
  <fct> <int>    <dbl>  <dbl>    <dbl>  <dbl>
1 F        47     2.36  0.274     9.20   1.36
2 M        97     2.9   0.467    11.3    2.54
```

```
ggplot(cats_df, aes(Bwt, Hwt, colour = Sex)) +
  geom_point(alpha = 0.75) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "Body mass (kg)",
    y = "Heart mass (g)",
    colour = "Sex"
  )
```




```
# Add diagnostics for the interaction model
q6_diagnostics <- augment(q6_interaction) |>
  mutate(
    .obs = row_number(),
    .abs_std_resid = abs(.std.resid)
  )

# Leverage and Cook's distance
q6_diagnostics |>
  slice_max(.hat, n = 5) |>
  select(.obs, Sex, Bwt, Hwt, .hat, .cooksd, .std.resid)

# Studentised residuals
q6_diagnostics |>
  slice_max(.abs_std_resid, n = 5) |>
  select(.obs, Sex, Bwt, Hwt, .hat, .cooksd, .std.resid)
```

```
# A tibble: 5 × 7
   .obs Sex     Bwt   Hwt   .hat .cooksd .std.resid
  <int> <fct> <dbl> <dbl>  <dbl>   <dbl>      <dbl>
1    97 M       3.9    20 0.0588  0.0580       3.62
2    48 F       3.7    11 0.0759  0.0408      -2.73
3    49 F       3.7    11 0.0759  0.0408      -2.73
4    50 F       3.7    11 0.0759  0.0408      -2.73
5    51 F       3.7    11 0.0759  0.0408      -2.73
```

```
# A tibble: 5 × 7
   .obs Sex     Bwt   Hwt   .hat .cooksd .std.resid
  <int> <fct> <dbl> <dbl>  <dbl>   <dbl>      <dbl>
1    97 M       3.9    20 0.0588  0.0580       3.62
2    48 F       3.7    11 0.0759  0.0408      -2.73
3    49 F       3.7    11 0.0759  0.0408      -2.73
4    50 F       3.7    11 0.0759  0.0408      -2.73
5    51 F       3.7    11 0.0759  0.0408      -2.73
```

The residual plots assess linearity, constant variance, normality, and influence. Variance-inflation factors quantify redundancy among the three model columns. Because an uncentred, continuous predictor appears in an interaction, some non-essential inflation is expected; centring body mass would reduce this without changing fitted values or the interaction test. Cook's distance flags observations for checking and sensitivity analysis, not automatic deletion.

## Question 7: Theophylline concentration profiles (/25)

The built-in `Theoph` dataset contains serum theophylline concentrations (`conc`, mg $L^{-1}$) measured repeatedly over approximately 25 hours after oral administration to 12 subjects. It also records subject body mass (`Wt`, kg), dose (`Dose`, mg $kg^{-1}$), and elapsed time (`Time`, h). The object has a specialised grouped-data class rather than being an ordinary tibble.

a. Convert `Theoph` to a plain tibble, convert `Subject` to an unordered factor, and arrange observations by subject and time. Audit the resulting data by reporting its dimensions, missing values, observations per subject, and whether `Wt` and `Dose` remain constant within each subject. Explain why these checks are necessary before calculating subject-level quantities. **(/5)**

b. Create a new dataset containing exactly one row per subject and the following variables:

- body mass and dose;
- number of concentration measurements;
- observed peak concentration;
- time at which the observed peak occurred; and
- area under the concentration-time curve from the first to last observation, calculated by joining successive observations with straight lines and summing the resulting trapezoid areas.

Present this dataset as a clearly labelled and appropriately rounded `gt` table. **(/7)**

c. Produce one figure that shows all subject-specific concentration profiles as lines and points. Superimpose a population summary obtained by dividing elapsed time into the intervals 0–0.5, >0.5–1, >1–2, >2–4, >4–8, >8–12, and >12–25 hours, then calculating the median and interquartile range of concentration within each interval. The figure must distinguish individual trajectories from the population summary and have a self-contained caption. **(/6)**

d. Determine whether body mass is associated with the subject-level area under the curve calculated in (b). Produce a labelled EDA figure, select and justify a suitable inferential measure, state the hypotheses, check the relevant assumptions, and report the estimate, 95% confidence interval, test statistic, degrees of freedom, and $p$-value. Interpret the evidence without treating a non-significant result as proof of no association. **(/7)**

Worked answer: Question 7

```
theoph <- as_tibble(datasets::Theoph) |>
  mutate(Subject = factor(Subject, ordered = FALSE)) |>
  arrange(Subject, Time)

dim(theoph)
```

```
[1] 132   5
```

```
colSums(is.na(theoph))
```

```
Subject      Wt    Dose    Time    conc
      0       0       0       0       0
```

```
q7_audit <- theoph |>
  summarise(
    observations = n(),
    distinct_weights = n_distinct(Wt),
    distinct_doses = n_distinct(Dose),
    .by = Subject
  )

q7_audit
```

```
# A tibble: 12 × 4
   Subject observations distinct_weights distinct_doses
   <fct>          <int>            <int>          <int>
 1 6                 11                1              1
 2 7                 11                1              1
 3 8                 11                1              1
 4 11                11                1              1
 5 3                 11                1              1
 6 2                 11                1              1
 7 4                 11                1              1
 8 9                 11                1              1
 9 12                11                1              1
10 10                11                1              1
11 1                 11                1              1
12 5                 11                1              1
```

```
q7_auc <- theoph |>
  summarise(
    auc = sum(diff(Time) * (head(conc, -1) + tail(conc, -1)) / 2),
    .by = Subject
  )

theoph_subject <- theoph |>
  distinct(Subject, Wt, Dose) |>
  left_join(q7_auc, by = "Subject")

cor.test(theoph_subject$Wt, theoph_subject$auc)
```

```

	Pearson's product-moment correlation

data:  theoph_subject$Wt and theoph_subject$auc
t = -1.1202, df = 10, p-value = 0.2879
alternative hypothesis: true correlation is not equal to 0
```

The subject is the independent observational unit. The hypotheses are $H_0 : \rho = 0$ and $H_A : \rho \neq 0$. The scatter plot is reasonably compatible with a straight-line association, neither variable shows strong evidence against normality, and no single point obviously determines the pattern, so Pearson's correlation is defensible. The estimated association is moderate and negative, but its confidence interval spans both substantial negative and modest positive values ($r = -0.335$, 95% CI $[-0.762, 0.296]$, $t_{10} = -1.12$, $p = 0.288$). The data therefore do not

## Question 8: Exercise and resting pulse (/25)

The `survey` dataset in **MASS** contains questionnaire responses from 237 university students. Relevant variables are sex (`Sex`), exercise frequency (`Exer`), resting pulse rate (`Pulse`, beats $min^{-1}$), and age (`Age`, years). This question compares students who reported exercising frequently with those who reported no exercise; students in the intermediate `"Some"` category are outside the target comparison.

a. Starting from `MASS::survey`, retain only `Sex`, `Exer`, `Pulse`, and `Age`. Keep the `"Freq"` and `"None"` exercise categories, remove incomplete rows, and:

- recode exercise as an ordered factor labelled `"None"` and `"Frequently"`;
- create age bands `"16–18"`, `"19–21"`, and `"22+"`; and
- report, in a compact audit, the original number of rows, the number excluded as `"Some"`, the number removed for incomplete required values, and the final number analysed.

Show enough checks to demonstrate that the resulting variables and category order are correct. **(/6)**

b. Calculate sample size, mean, median, standard deviation, and IQR of pulse rate for each exercise-by-sex combination. Present the result as a polished `gt` table with units and sensible rounding. **(/5)**

c. Construct one figure comparing the pulse-rate distributions between exercise groups. It must show the distribution shape, a compact quartile summary, individual observations coloured by age band, separate panels for sex, and the mean with a 95% confidence interval. Use position adjustments that keep all layers readable and provide a self-contained caption. **(/7)**

d. Ignoring sex for this specified unadjusted comparison, determine whether mean pulse rate differs between the two exercise groups. State the observational unit and hypotheses, use graphical and formal evidence to assess the relevant assumptions, select and conduct an appropriate analysis, and report the estimated difference with its 95% confidence interval, test statistic, degrees of freedom, and $p$-value. Discuss one reason why this observational comparison should not be interpreted causally. **(/7)**

Worked answer: Question 8

```
survey_raw <- as_tibble(MASS::survey)

q8_original_n <- nrow(survey_raw)
q8_some_n <- sum(survey_raw$Exer == "Some", na.rm = TRUE)

survey_target <- survey_raw |>
  dplyr::select(Sex, Exer, Pulse, Age) |>
  filter(Exer %in% c("Freq", "None"))

q8_incomplete_n <- sum(!complete.cases(survey_target))

survey_clean <- survey_target |>
  drop_na() |>
  mutate(
    exercise = factor(
      Exer,
      levels = c("None", "Freq"),
      labels = c("None", "Frequently"),
      ordered = TRUE
    ),
    age_band = cut(
      Age,
      breaks = c(-Inf, 19, 22, Inf),
      right = FALSE,
      labels = c("16–18", "19–21", "22+")
    )
  ) |>
  dplyr::select(-Exer)

tibble(
  stage = c(
    "Original rows",
    "Excluded: Some exercise",
    "Removed: incomplete required values",
    "Final analysed rows"
  ),
  n = c(
    q8_original_n,
    q8_some_n,
    q8_incomplete_n,
    nrow(survey_clean)
  )
)
```

```
# A tibble: 4 × 2
  stage                                n
  <chr>                            <int>
1 Original rows                      237
2 Excluded: Some exercise             19
3 Removed: incomplete required values 29
4 Final analysed rows                189
```

```
t2_q8 <- t.test(Pulse ~ exercise, data = survey_clean, var.equal = FALSE)
t2_q8
```

```

	Welch Two Sample t-test

data:  Pulse by exercise
t = 2.2986, df = 99.082, p-value = 0.02386
alternative hypothesis: true difference in means between group None and group
```

The student is the observational unit. With the factor ordered as `"None"` then `"Frequently"`, the estimated contrast is mean pulse in the no-exercise group minus mean pulse in the frequent-exercise group. The hypotheses are $H_0 : \mu_{\text{None}} - \mu_{\text{Frequently}} = 0$ and $H_A : \mu_{\text{None}} - \mu_{\text{Frequently}} \neq 0$. The Q-Q plots and within-group tests do not show strong departures from normality. Welch's two-sample procedure is a cautious choice because it does not require equal population variances and accommodates the unequal sample sizes.

## Coverage and marking map

| Question | Main assessed work | Chapters | Marks |
|---|---|---|---|
| 1 | Data structure, descriptive statistics, visualisation, distributions, transformation, sampling uncertainty, EDA | 1–6 | 20 |
| 2 | Hypotheses, assumptions, paired *t*-test, estimation and interpretation | 5–7, 10 | 16 |
| 3 | Group EDA, one-way ANOVA alternatives, residual diagnostics, multiplicity-controlled comparisons | 5–6, 8, 10–11 | 20 |
| 4 | Correlation, simple regression, diagnostics, confidence and prediction intervals | 9, 11–12 | 22 |
| 5 | Linear versus polynomial regression, nested comparison, diagnostics and bounded interpretation | 11, 13 | 18 |
| 6 | Multiple regression, categorical predictors, nested models, diagnos- | 11, 14 | 24 |

| Question | Main assessed work | Chapters | Marks |
|---|---|---|---|
| | tics, collinearity and influence | | |
| 7 | Longitudinal-data wrangling, subject-level integration, layered visualisation, association | 2–5, 9–10 | 25 |
| 8 | Filtering, recoding, grouped summaries, layered faceted visualisation, two-group inference | 2–3, 5–7, 10 | 25 |
| **Total** | | **1–14** | **170** |

# Bibliography